SCOPED STUDY | BASELINE: 2 MiB L2

CLAIM UNDER REVIEW At least one larger L2 reduces 99th-percentile frame time by removing capacity misses while meeting the declared power, area, and timing checks.

Intent

Reduce XR tail latency without weakening power, area, timing, or functional checks.

Task

Decide whether the 3 MiB or 4 MiB L2 should advance beyond the rerun 2 MiB baseline.

Design space

2 MiB, 3 MiB, and 4 MiB L2 capacities; all other architecture choices remain fixed.

Recorded study state

XR trace, frame IDs, cache configuration, software image, power model, and tool versions.

Environment

Cycle-level simulator and SRAM model under the declared workload and process conditions.

Method role

Directly enumerate all three legal capacities; no learned method is required.

Feedback budget

Four cycle-level executions, including reruns, plus one SRAM evaluation per capacity.

Evidence

Checks, misses, power, area, access time, and uncertainty in the 99th-percentile frame-time change.

Failed runs / rejected alternatives

Links distinguish infrastructure failures from candidates rejected by a declared check.

Rejection checks and authority

Functional, power, area, timing, and tail-latency checks may reject; the architect owns waivers.

What the evidence supports

Open until valid returns pass checks and the capacity-miss mechanism explains tail latency.

Accountable decision

The named architect may advance at most one capacity to RTL evaluation; no signoff or product commitment.

This optional card indexes the study record. It does not replace existing engineering records.